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Processando config 23, seed 4, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

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HOST = matheus-IdeaPad-3-15ALC6, Linux

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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

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smultin_MultinomialBitsOver test:

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N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE

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Number of bits = n = 8388608
Number of cells = 2^L = 274877906944
Expected number per cell = 1 / 32768
EColl = n^2 / (2k) = 128
Hashing = TRUE

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Collision test
CollisionOver: density = n / k = 1 / 32768.00
Expected number of collisions = Mu = 128.00

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Results of CollisionOver test:

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POISSON approximation :
Expected number of collisions = N*Mu : 256.00
Observed number of collisions : 295
p-value of test : 9.1e-3

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Total number of cells containing j balls

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j = 0 : 549739036967
j = 1 : 16776626
j = 2 : 295
j = 3 : 0
j = 4 : 0
j = 5 : 0

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CPU time used : 00:00:04.50

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Generator state:
16777216 bits have been read.

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HOST = matheus-IdeaPad-3-15ALC6, Linux

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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 17.00
p-value of test : 0.39

CPU time used : 00:00:00.04

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.08

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 0.21
p-value of test : 0.42

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 6.47
p-value of test : 0.49

Normal statistic for number of jumps : -51.35
p-value of test : 1 - eps1 *****

CPU time used : 00:00:00.05

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 0.60
p-value of test : 0.28

CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -0.092
p-value of test : 0.54

CPU time used : 00:00:00.20

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.015
p-value of test : 0.79

Kolmogorov-Smirnov- statistic = D- : 0.029
p-value of test : 0.40

Anderson-Darling statistic = A2 : 0.60
p-value of test : 0.65

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 3.31
p-value of test : 0.35

Global longest run of 1 : 22.00
p-value of test : 0.63

CPU time used : 00:00:00.07

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 105.97
p-value of test : 0.19

CPU time used : 00:00:00.00

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 16.71
p-value of test : 0.78

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.90
p-value of test : 0.18

CPU time used : 00:00:00.01

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -0.23
p-value of test : 0.59

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 0.53
p-value of test : 0.30

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 201.87
p-value of test : 0.10

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 272.40
p-value of test : 0.91

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 469.74
p-value of test : 0.73

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

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-----  
Normal statistic          :   -0.21  
p-value of test          :    0.58  
-----
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CPU time used             :   00:00:00.01
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Generator state:  
269499904 bits have been read.
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin:   /home/matheus/Área de  
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin
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sstring_AutoCor test:
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-----  
N = 1,  n = 16777154,  r = 0,  s = 32,  d = 2  
-----
```

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-----  
Normal statistic          :    0.040  
p-value of test          :    0.48  
-----
```

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CPU time used             :   00:00:00.00
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Generator state:  
286277088 bits have been read.
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin:   /home/matheus/Área de  
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin
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sstring_Run test:
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-----  
N = 1,  n = 3355443,  r = 0,  s = 32  
-----
```

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-----  
Total number of 1 runs: 3355446
```

```
Number of degrees of freedom :   36  
Chi2 statistic for number of runs : 22.23  
p-value of test              :    0.96  
-----
```

```
-----  
Total number of bits: 13422368
```

```
Normal statistic for number of bits :    0.12  
p-value of test                      :    0.45
```

CPU time used : 00:00:00.05

Generator state:
299699456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 1.43
p-value of test : 0.70

CPU time used : 00:00:00.07

Generator state:
316476672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:
333167872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	42.10
p-value of test	:	0.47

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	40.39
p-value of test	:	0.54

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	46.20
p-value of test	:	0.95

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	20.46
p-value of test	:	0.98

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	16.75
p-value of test	:	0.78

CPU time used : 00:00:00.04

Generator state:
349945088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	91.84
p-value of test	:	0.54

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	94.72
p-value of test	:	0.46

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	499.59
p-value of test	:	0.64

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	100.86
p-value of test	:	0.20

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	58.73
p-value of test	:	0.19

CPU time used : 00:00:00.04

Generator state:

366722304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	145.68
p-value of test	:	0.16

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	173.38
p-value of test	:	7.8e-3

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	160.84
p-value of test	:	0.49

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	119.19
p-value of test	:	0.74

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	86.04
p-value of test	:	0.57

CPU time used : 00:00:00.02

Generator state:

383499104 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:05.98
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test		p-value

5	LinearComp	1 - eps1
22	MatrixRank, 320 x 320	eps

All other tests were passed